

Giardia lamblia ‘Grand oldman of intestine’- a human intestine dwelling flagellate.

Unit

5

Locomotion and Reproduction

... In search of prey and perpetuation

LOCOMOTION

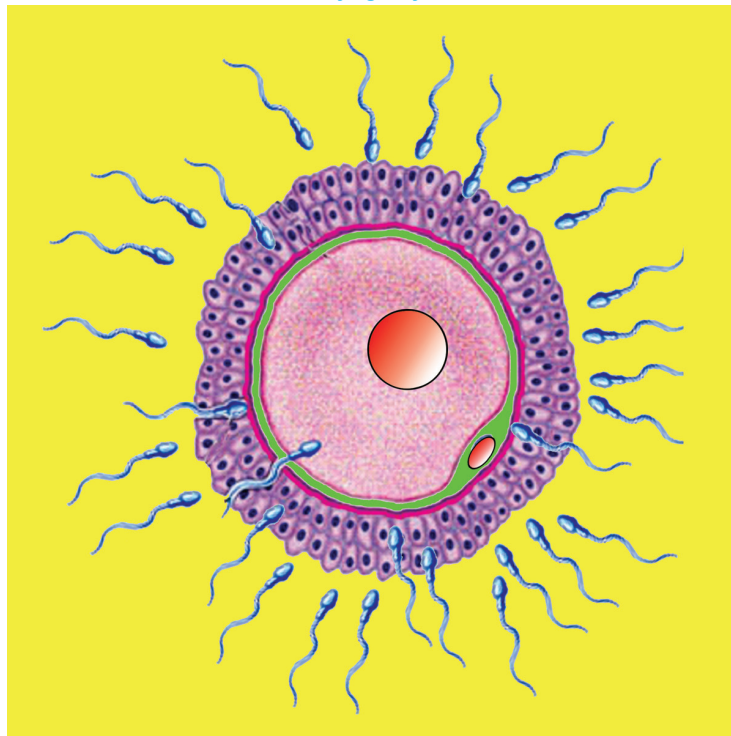
In our day-to-day life we move our limbs, jaws, eyelids, tongue, etc. without any change in the position or location of our body. Plants perform tropic movements in response to stimuli of light and gravity. These movements do not result in any change in the place or location of the body of an organism.

Locomotion is defined as the voluntary movement of an organism from one place to another either in search of food or shelter or mate or to escape from the predators. Walking, running, climbing, swimming, crawling, flying, etc., are the different forms of locomotory movements exhibited by various types of animals. Locomotory structures may also be used for the collection of food. For example, protists such as Amoeba and Paramecium use their locomotor structures (pseudopodia and cilia, respectively) for the collection of food also. The above facts suggest that movements and locomotion cannot be studied separately; they may be linked by stating that “all locomotions are movements, but all movements are not locomotions”. The method of locomotion performed by animals varies with their habitats and the demand of the situation.

REPRODUCTION

It is defined as a biological process in which an organism gives rise to young ones (offspring) similar to itself (perpetuation of the race) The offspring grow, mature into adults and in turn produce their offspring. Thus, there is a cycle of birth, growth and death. Reproduction enables the continuity of the species, generation after generation. There is a large diversity in the biological world and each organism has evolved its own mechanism to multiply and produce offspring. The organism's habitat, its internal physiology and several other factors are collectively responsible for its mode of reproduction. Based on the participation of one organism or two in the process of reproduction, it is of two types, asexual and sexual.

Syngamy



(Sperm meets the ovum by its lashing movements)

Locomotion and Reproduction in Protozoa

- 5.1 Locomotion in Protozoa
- 5.2 Flagellar and ciliary movement
- 5.3 Asexual reproduction in Protozoa
- 5.4 Sexual reproduction in protozoa

5.1 Locomotion in Protozoa

Though the protozoans are acellular and mostly microscopic, they are versatile in their locomotion. The free living aquatic protozoans move through the water, whereas some parasitic forms move through the body fluids of their hosts. Locomotion in protozoans is performed by cellular extensions such as **pseudopodia**, short hair-like organelles, the **cilia**, whip-like organelles, the **flagella** and the contractile fibrils, the **myonemes**.

5.1.1 Pseudopodia

These are found in **rhizopods**. The pseudopodia are temporary extensions of cytoplasm that develop in the direction of the movement. These temporary structures are useful to move on the substratum as our legs do, hence the name 'pseudopodia'. There are four kinds of pseudopodia namely **lobopodia** (blunt finger-like) as in *Amoeba* and *Entamoeba*, **filopodia** (fibre-like) as in *Euglypha*, **reticulopodia** (net-like) as in *Elphidium* and **axopodia** or **heliopodia** (sun ray-like) as in *Actinophrys*.

Pseudopodium is formed by the conversion of gel (viscous outer endoplasmic part) to sol (fluid-like inner endoplasmic part) and vice-versa. There are different theories on the formation of pseudopodium. The sol-gel transformation theory is the most accepted theory. Opinions differ on the site of contraction that pushes the pseudopodium forwards. Allen's theory of 'Front contraction' or 'Fountain zone' theory is

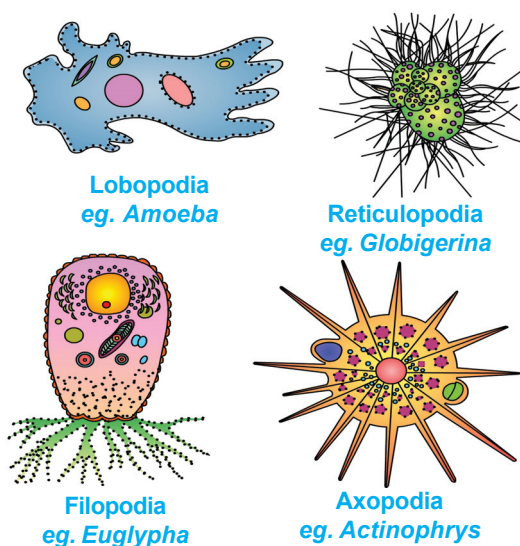


Fig. 5.1 Types of Pseudopodia

Plasmagel $\xrightarrow{+H_2O(\text{Solation})}$ Plasmasol $\xrightarrow{-H_2O(\text{Gelation})}$ Pseudopodium $\longrightarrow$ Displacement of the body in the forward direction

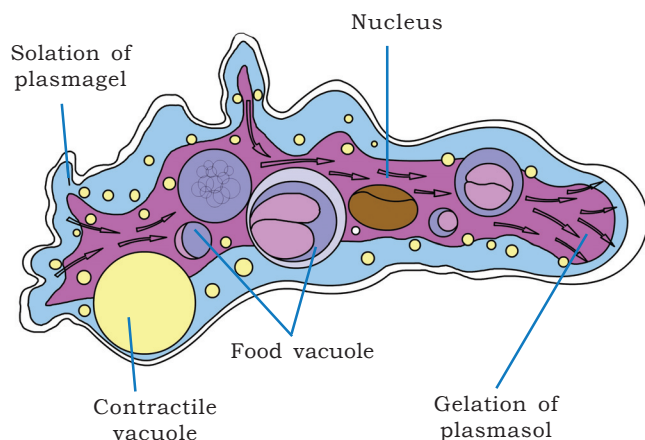


Fig. 5.2 Amoeba-locomotion

considered to be more appropriate. However modern research brings in the role of actin and myosin molecules too.

The pseudopodial or amoeboid movement is performed by **Amoeba**, **Entamoeba**, **Polystomella**, **Actinophrys**, etc. Amoeboid locomotion is probably the most primitive and the slowest type of locomotion.

In addition to protozoans, amoeboid locomotion is performed by amoeboid cells, macrophages, neutrophils, etc., of higher metazoans.

5.1.2 Flagella

The long whip-like locomotor organelles are called flagella (sing. flagellum). These are found in the **mastigophoran protozoans** (mastig – whip; phoran – bearer). Bacteria also possess flagella, which are structurally different from those of the eukaryotes. The movement of spermatozoa of animals involves flagellar movement.

I. Structure

The structural components of a typical flagellum include axoneme, microtubules, dynein arms, inner sheath, outer sheath, radial spokes, lateral appendages (such as mastigonemes or flimmers) and a basal granule (kinetosome).

- i. **Axoneme/ axial filament:** It is the central, longitudinal, microtubular structure of cilium and flagellum. It is surrounded by a membrane which is continuous with plasma membrane. All the components of the axoneme are embedded within the matrix.
- ii. **Microtubules:** An axoneme is made up of 2 central 'singlets' and 9 peripheral 'doublets (9+2 array)'. These are formed by the protein called **tubulin**. Each peripheral doublet consists of an outer 'A' and the inner 'B' tubules. So, the peripheral tubules are actually 9 microtubular doublets. (The microtubule 'A' is smaller but complete whereas 'B' is larger and incomplete). The peripheral doublets are interconnected by 'linkers' called **nexins**.
- iii. **Dynein arms:** The 'A' tubule of each peripheral doublet bears paired arms along its length, called dynein arms (Dyne - pulling like a dynamo). The dynein arms of 'A' tubule face the tubule 'B' of the adjacent doublet. They are oriented in the

same direction (clockwise) in all microtubules. The dynein arms are considered '**protein motor molecules**'. They are made of the protein, **dynein**.

- iv. **Inner and outer sheaths:** The two central singlets are enclosed by a fibrous inner sheath and the peripheral doublets are enclosed by an outer sheath (an extension of plasma membrane). The central singlets do not reach below the level of the pellicle or plasmalemma.
- v. **Radial spokes:** They are elastic fibres that connect the inner sheath, with the 'A' tubule of each doublet. They resemble the spokes that connect the rim of the wheel of a bicycle with the centre, hence the name 'radial spokes/ radial bridges'. The nine radial spokes limit the extent of sliding past of the doublets, during bending movements.
- vi. **Basal granule / kinetosome:** It is the cell organelle involved in the formation of a flagellum / cilium. The basal granule is a modified centriole. It is also called **kinetosome** [kineto - moving; soma - body] or **basal body** or **blepharoplast**. It is present in the ectoplasm. The basal granule is a cylindrical body, formed by '9' peripheral triplets arranged in a circle. The tubules of each triplet are designated A, B and C from the center towards the periphery. Both 'A' and 'B' cross the basal

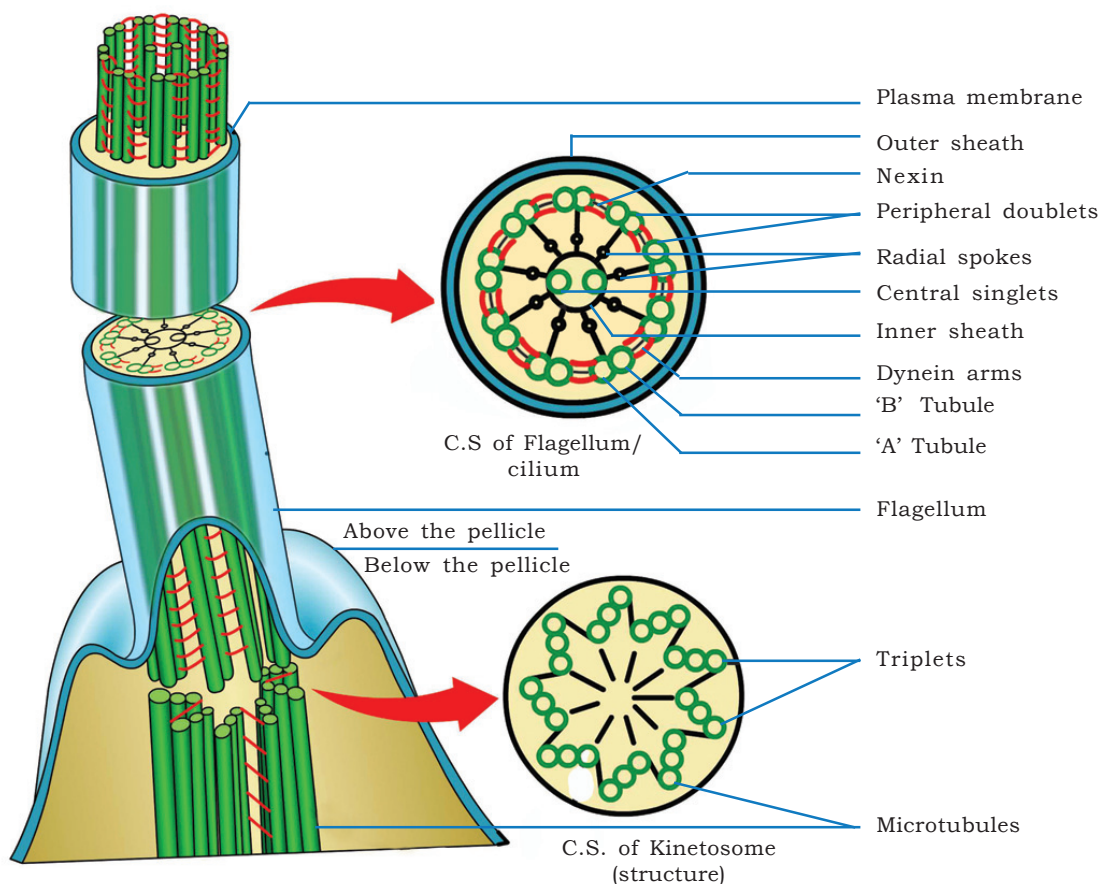


Fig. 5.3 Structure of a flagellum/ cilium

plate and continue as peripheral doublet above the pellicle in the axoneme. However, the tubule C terminates at the basal plate. Thus the 'triplets' of the basal granule are converted into the flagellar/ciliary 'doublets'. There are no central microtubules in the basal granule. The basal granule is also connected to the plasma membrane and nucleus by communication tubules, called **rootlets**. The rootlets can pull the flagellum and alter its orientation.

- vii. **Lateral appendages:** Some flagella bear one or two or many rows of short, lateral hair like fibrils called lateral appendages, also called '**mastigonemes**'.

II. Types of Flagella

Based on the presence or absence and/or the number of rows of lateral appendages, five types of flagella are recognised.

- Stichonematic:** This flagellum bears one row of lateral appendages on the axoneme. E.g. *Euglena* and *Astasia*.
- Pantonematic:** This flagellum has two or more rows of lateral appendages on the axoneme. E.g. *Peranema* and *Monas*.
- Acronematic:** This type of flagellum does not bear lateral appendages and the terminal part of the axoneme is naked without the outer sheath at its tip. E.g.: *Chlamydomonas* and *Polytoma*.

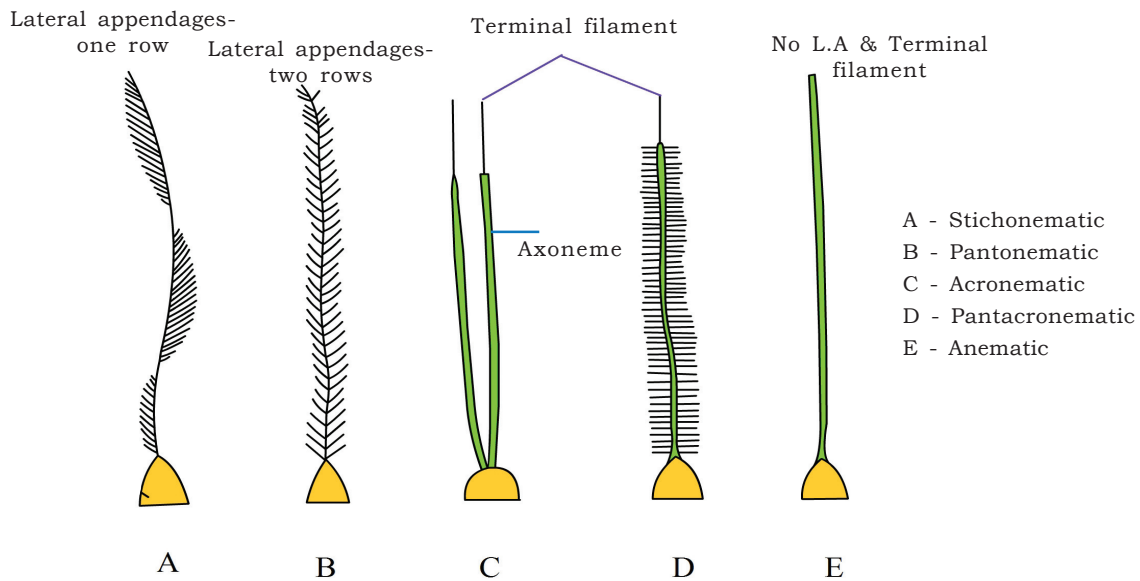


Fig. 5.4 Types of flagella

- d. **Pantacronematic:** This type of flagellum is provided with two or more rows of lateral appendages and the axoneme ends in a terminal naked filament. E.g. *Urceolus*.
- e. **Anematic or simple:** In this type of flagellum, lateral appendages and terminal filament are absent. Hence, it is called anematic (a-no; nematic-threads)
e.g. *Chilomonas* and *Cryptomonas*

III. Number of flagella

In the mastigophorans, the number of flagella varies from species to species. *Euglena* has two flagella (one long and one short), *Trypanosoma* has a single long flagellum arising at the rear end of the body. Four flagella are found in *Trichomonas*, four pairs in *Giardia* and many in *Trichonympha*.

5.1.3 Cilia

I. Structure

Cilia are small hair-like structures found in ciliate protists, epithelial lining of respiratory tract, genital ducts, ventricles of brain etc. In the primitive ciliates like *Paramecium*, the entire body is covered by cilia and in the advanced ciliates, such as *Vorticella*, cilia are restricted to the peristomial region. In the suctorians such as *Acineta*, the cilia are confined to the 'juvenile stages' only (The adults possess suctorial tentacles which help in feeding).

Cilia arise from the basal granules situated in the ectoplasm below the pellicle. Cilia are similar to flagella in their ultra structure. Cilia serve as organelles of locomotion, food collection and also act as sensory structures.

II. Infraciliary system

It is located just below the pellicle in the ectoplasm of a ciliate. It includes **kinetosomes**, **kinetodesmal fibrils** and **kinetodesmata**. The kinetosomes are present at the bases of cilia in transverse and longitudinal rows. The kinetodesmal fibrils are connected to the kinetosomes and run along the right side of each row of kinetosomes as a 'cord of fibres' called **kinetodesmata**. A longitudinal row of kinetosomes together with kinetodesmata constitute a unit called '**kinety**'.

All the kineties together form an infraciliary system, which is connected to a '**motorium**', located near the cytopharynx. The infraciliary system and motorium form the 'neuromotor system' that controls and coordinates the movement of cilia.

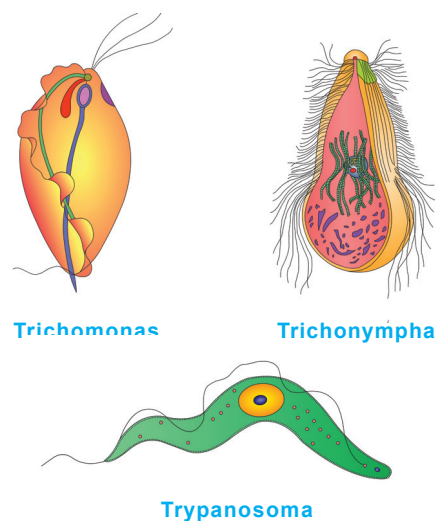


Fig. 5.5 Flagellates

III. Differences between Flagella & Cilia

Flagella	Cilia
1) Long whip-like	1) Short hair-like
2) One, two, four or many	2) Many
3) Flagella normally originate at the anterior end.	3) Cilia are found all over the body surface or confined to peristomial region.
4) Flagella do not unite to form cirri.	4) In some ciliates, the cilia fuse to produce cirri, or undulating membrane and membranelles,
5) Flagella perform “undular” movement	5) Cilia perform “pendular” movement

5.1.4 Myonemes

They are the contractile fibrils present below the pellicle in the ectoplasm. They occur in flagellates, apicomplexans and ciliates.

5.2 FLAGELLAR AND CILIARY MOVEMENT

5.2.1 Swimming locomotion

This type of locomotion in protozoans is brought about by **flagella** and **cilia**. Flagella and cilia are also called ‘**undulipodia**’ by Hyman. They are useful in bringing about locomotion in water and the body fluid of hosts in the case of a few parasites. Bending movement of a flagellum is brought about by the sliding of microtubules past each other due to the functioning of ‘**dynein arms**’ utilising ATP. A flagellum pushes the

fluid medium at right angles to the point of its attachment, by its bending movement.

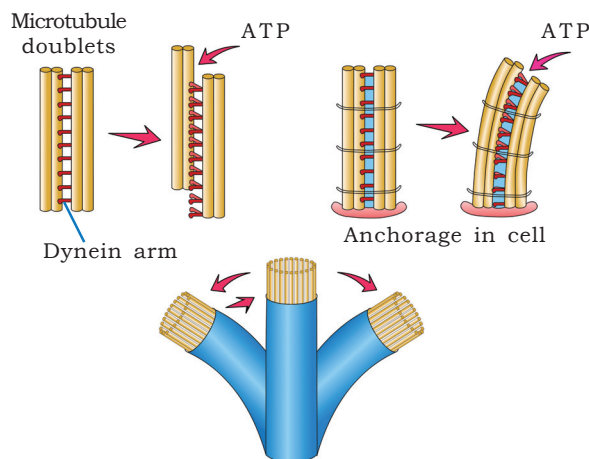


Fig 5.6 Bending of microtubules

Bending movement of flagella and cilia

Dynein arms show a complex cycle of movements using energy provided by ATP (dynein arms are the sites of ATPase activity in the cilia and flagella). The dynein arms of each doublet attach to an adjacent doublet *and pull* the neighbouring doublet. So the doublets slide past each other in opposite directions. The arms release and reattach a little farther on the adjacent doublet and again ‘pull’. As the doublets of a

flagellum or cilium are physically held in place by the radial spokes, the doublets cannot slide past much. Instead, they curve and cause bending of flagellum or cilium. Such bending movements of flagella and cilia play an important role in the flagellar and ciliary locomotion.

If, nexins and radial spokes of an axoneme in a flagellum/cilium are subjected to enzymatic digestion and provided with ATP, what happens to the doublets?

5.2.2 Flagellar locomotion

Flagellum shows undulation movements, sidewise lash movements and spiral gyration movements.

- a) **Undulation movements:** Undulatory movements from the base to the tip causes a **pushing force** (like the propeller of a boat) and the organism is *pushed* backwards (undular movement). Undulation from the tip to the base causes a **pulling force** (like the propeller of an aeroplane) and the organism is *pulled* forwards. If the flagellum bends to one side and shows a wave like movement from the base to the tip, the organism moves laterally in the opposite direction.
- b) **Sidewise-lash movement:** The flagellar movement of many organisms is a sidewise-lash which consists of two 'strokes' namely the **effective** or **propulsive stroke** and the **recovery stroke**.
 - (i) **Effective stroke:** Flagellum becomes rigid and starts bending to one side beating against the water. This beating against water is at right angles to the body axis and the organism moves forwards (the direction of movement is parallel to the surface of the body).
 - (ii) **Recovery stroke:** Flagellum becomes comparatively soft so as to offer least resistance to water and moves backwards to its original position. It is called 'recovery stroke'.
- c) **Simple conical gyration movements:** In simple conical gyration movement, a flagellum turns like a screw. This exerts the propelling action that pulls the organism forwards through water with a spiral **rotation** around the axis of movement and **gyration** on its own axis.

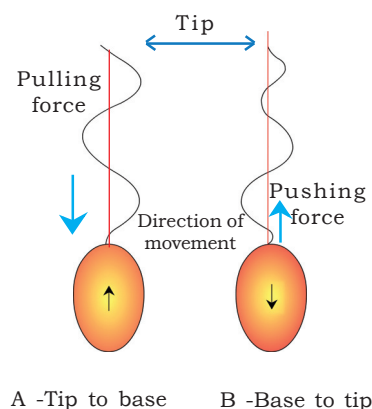


Fig. 5.7 Undular movement of a flagellum

5.2.3 Ciliary Locomotion

Ciliary locomotion is observed in ciliates. It is performed by bending movements of the cilia. Cilium also shows swift back and forth movements during locomotion as in the case of a flagellum. These are called *effective* and *recovery* strokes respectively. Cilium moves the water parallel to the surface of its attachment like that of a *paddle* or *pendular movement*. The movement of water may also be described as perpendicular to the axis of cilium.

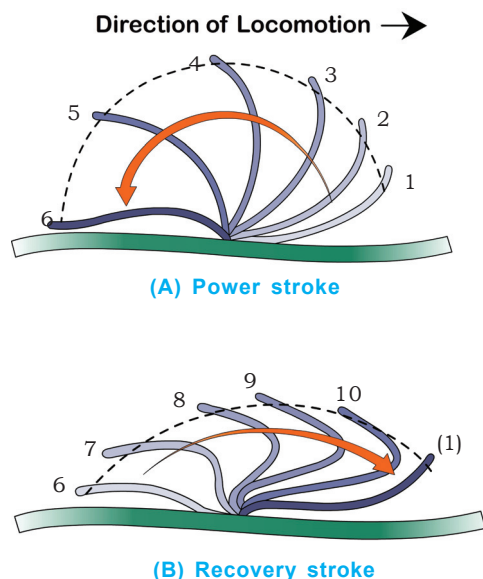


Fig. 5.8 (A) Effective stroke
(B) Recovery stroke

- a) **Effective stroke:** Cilium bends backwards and beats against the water. It is called the 'effective stroke'. Body moves forwards in this stroke, as cilium pushes the water backwards. Cilium moves like *pendulum* or a *paddle*. (*pendular movement*)
- b) **Recovery stroke:** The cilium after the effective stroke, comes to its original position by its backward movement without any resistance. It is called 'recovery stroke'.

In **Paramecium** cilia show two types of coordinated movements.

- a. **Synchronous movement:** Cilia in a transverse row beat simultaneously in one direction. It is called **synchronous movement**.
- b. **Metachronous movement:** The sequential movement of cilia, in a longitudinal row, one after the other in one direction is called **metachronous**

movement. This movement passes like a wave. It resembles the movement of plants in a paddy field due to the blow of wind from one direction to the other. The ciliary locomotion is coordinated by the *infraciliary system* through a neuromotor centre called **motorium**, present near the cytopharynx. Ciliary locomotion is faster than flagellar locomotion. Hence, ciliates are the fastest protozoans.

Flagella are responsible for the movement of the organism, whereas cilia bring out movement of either the organism or the surrounding fluid as seen in ciliary mode of feeding in certain organisms.

5.2.4 Gliding Locomotion

The small zig-zag movement in some protozoans caused by the contraction and relaxation of myonemes present below the pellicle in the ectoplasm is called **gliding locomotion**. **Myonemes** are contractile fibrils which are similar to the myofibrils of higher protozoans. e.g. ciliates, flagellates etc.

Is it correct to say 'all undulipodia' do not show undular movements?

5.3 ASEXUAL REPRODUCTION

When an offspring is produced by a single parent without the involvement of gamete formation, the reproduction is **asexual**. In this method, a single individual (parent) is capable of producing the offspring. As a result, the offspring are not only identical to one another but also exact copies of their parent. The term '**clone**' is used to describe such morphologically and genetically similar individuals. Asexual reproduction is widespread among different organisms. It is common in members of Protista, Bacteria, Archaea and in multicellular organisms with relatively simple organisation. The offspring show '**uniparental inheritance**' without any genetic variations.

5.3.1 Asexual Reproduction in Protozoans

The methods of asexual reproduction that occur in many protozoans are binary fission and multiple fission.

5.3.2 Binary Fission

It is the most common method of asexual reproduction in protozoans. It occurs during **favourable conditions**. It is the division of the parent cell into two daughter individuals. It involves the division of nucleus (karyokinesis) followed by the division of the cytoplasm (cytokinesis). The plane of fission differs in different protozoans. Based on the axis of cytokinesis, the binary fission is chiefly classified into longitudinal and transverse fissions.

I. Longitudinal Binary Fission

It is seen in flagellates such as ***Euglena***, ***Trypanosoma*** etc. In this type of binary fission, the body divides into two halves longitudinally, hence called longitudinal **binary fission**.

Euglena possesses two flagella arising from two basal granules, one contractile vacuole, stigma, paraflagellar body (photoreceptor), cytostome, cytopharynx and reservoir anteriorly, nucleus in the centre and chromatophores around the nucleus and myonemes beneath the proteinous pellicle.

During the process of binary fission, the nucleus, basal granules, chromatophores, cytoplasm undergo division. The nucleus divides by **mitosis** into two daughter nuclei. Then the kinetosomes and the chromatophores also divide. At first, a longitudinal groove develops in the middle of the anterior end. This groove extends gradually towards the posterior end until

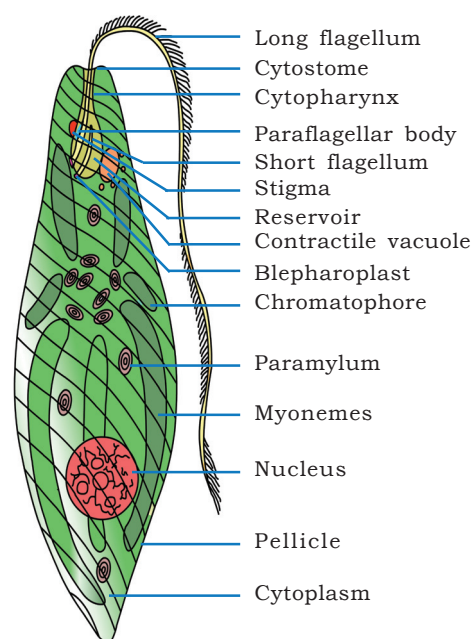


Fig. 5.9 *Euglena*

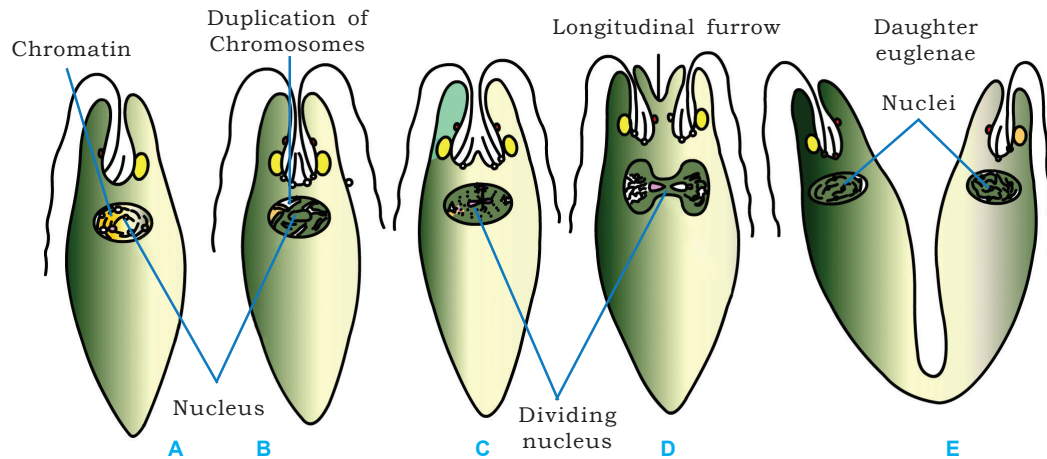


Fig. 5.7 Longitudinal binary fission in *Euglena* - different stages

the two daughter individuals are separated. One daughter *Euglena* retains the parental flagella. The other daughter individual develops new flagella from the newly formed basal granules. The stigma, paraflagellar body and contractile vacuole of the parent disappear. They develop afresh in both the daughter euglenae. **The longitudinal binary fission is known as symetrogenic division**, because the two daughter euglenae resemble each other like ‘**mirror images**’.

II. Transverse Binary Fission

It is performed by *Paramecium*, commonly called “slipper animalcule”. It has an oral groove, cytostome and cytopharynx in the oral surface. It also has one macronucleus (**polyploid**) and one micronucleus (**diploid**), two contractile vacuoles (anterior and posterior), trichocysts, infraciliary system and many cilia on the body.

During favourable conditions, *Paramecium* stops feeding after attaining its

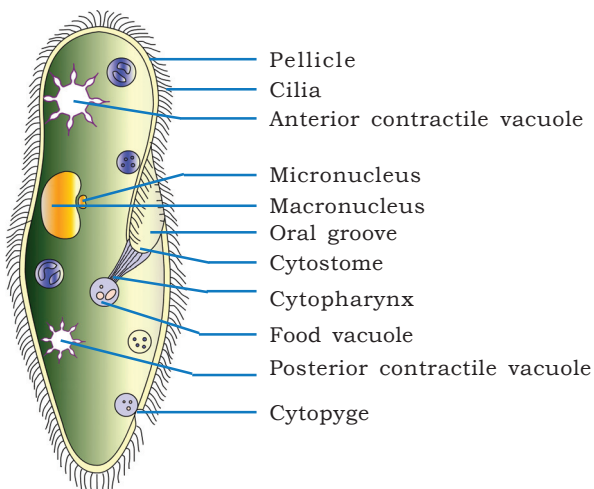


Fig 5.8 *Paramecium*

maximum growth. At first the micronucleus divides by **mitosis** and the macronucleus divides into two daughter nuclei by **amitosis**. The oral groove disappears. After karyokinesis, a transverse constriction appears in the middle of the body, which deepens and divides the parent cell into two daughter individuals, the anterior **proter** and the posterior **opisthe**. The proter receives the anterior contractile vacuole, cytopharynx and cytosome from its parent individual. It develops posterior contractile vacuole and a new oral groove. The **opisthe** receives the posterior contractile vacuole of its parent. It

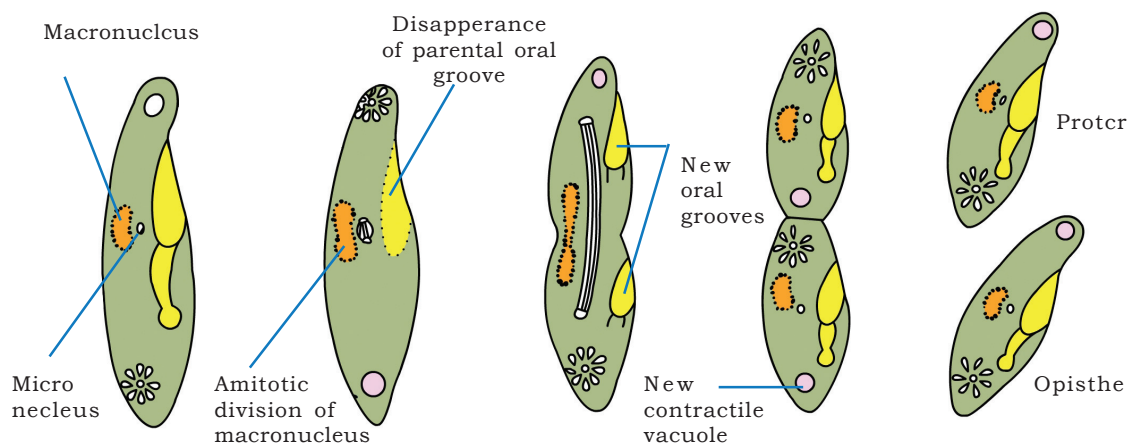


Fig. 5.9 Stages of transverse binary fission in *Paramecium*

develops a new anterior contractile vacuole, cytopharynx, cytostome and a new oral groove. Binary fission is completed in almost two hours, in favourable conditions and ***Paramecium*** can produce four generations of daughter individuals by binary fission in a day.

The transverse binary fission is also called **homothetogenic fission**, because the plane of fission is at right angles to the longitudinal axis of the body. As it occurs at right angles to the kineties, it is also called **perkinetal** fission (across the Kinetia).

5.3.3 Multiple Fission

It is the division of the parent body into many smaller individuals (Multi – many; Fission – splitting). Normally multiple fission occurs during **unfavourable** conditions. During the multiple fission, the nucleus first undergoes repeated mitotic divisions without cytokinesis. This causes the formation of many daughter nuclei. Then the cytoplasm also divides into as many number of bits as there are nuclei. Each cytoplasmic bit encircles one daughter nucleus. This results in the formation of many smaller individuals from a single parent organism.

There are different types of multiple fissions in protozoans such as **schizogony**, **male gametogony**, **sporogony** in ***Plasmodium***, **sporulation** in ***Amoeba***, etc.

5.4 SEXUAL REPRODUCTION

When two mature individuals (of opposite sex) participate in the reproductive process involving fusion of male and female gametes (**the products of meiosis** usually), it is called **sexual reproduction**. These haploid gametes fuse to form the diploid zygote, which develops into the new organism. It is an elaborate, complex and slow process compared to asexual reproduction. Because of the fusion of male and female gametes, sexual reproduction results in offspring that are not identical to the parents or amongst themselves.

5.4.1 Sexual reproduction in Protozoans

It takes place by the fusion of 'pronuclei' (haploid gametic nuclei) with or without the formation of gametes. Genetic recombination occurs in sexual reproduction during the formation of pronuclei or gametic nuclei (due to **meiotic crossovers**) and also due to the fusion of gametes from two different individuals. Sexual reproduction in protozoans occurs by syngamy. Conjugation, 'a process of nuclear reorganisation that occurs in some ciliates' also brings about a kind of sexual multiplication of the organism.

5.4.2 Syngamy

It is the fusion of two gametes.

- i) The fusion of similar gametes is called **isogamy** e.g. *Monocystis*.
- ii) The fusion of dissimilar gametes is called '**anisogamy**' e.g. *Plasmodium*.
- iii) **Hologamy** is the fusion of two mature organisms which do not form gametes but behave as gametes. e.g. *Trichonympha*.

* Union of pronuclei of the gametes is called '**amphimixis**' and the resultant nucleus is called **synkaryon**.

5.4.3 Conjugation

It is observed in ciliate protozoans such as *Paramecium* and *Vorticella*. Wichterman defined conjugation as a temporary union between **two senile** ciliates that belong to two different 'mating types' for the exchange of nuclear material and its reorganization. It restores the **vigour** and **vitality** which are lost due to chromosomal imbalance in their macronuclei caused by repeated **amitotic** divisions. Unfavourable conditions also induce conjugation.

GLOSSARY

Phototropism: It is the tendency of an organism to move towards or to move away from the source of light stimulus.

Organelle: A living component of the cell with a specific structure and function. e.g., Mitochondrion, Lysosome, Ribosome, Golgicomplex, etc.

Heliopodia: These are sunray-like pseudopodia present all over the surface of the body, as in 'sun animal cules' (*Actinophrys*, *Actionsphaerium*).

Tubulin: A protein with which the microtubules of cytoskeleton are formed.

Kinetosome: It is a modified centriole from which the cilium or flagellum is formed.

Suctoria: Advanced ciliate protozoans, which possess suctorial tentacles in the adult condition and cilia in the juvenile stages. e.g. *Acineta*.

Senile: Becoming old.

Nexin links: Proteins that connect the 'A' tubule of one doublet with the 'B' tubule of the next doublet in a clockwise manner in an axoneme are called nexin links. They maintain the integrity of the axoneme by holding doublets together.

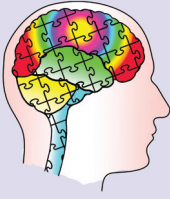
VERY SHORT ANSWER TYPE QUESTIONS

1. Draw a labelled diagram of T.S of flagellum.
2. List any two differences between a flagellum and a cilium.
3. What are dynein arms? What is their significance?
4. What is a kinety?
5. Distinguish between synchronous and metachronous movements.

6. Why do we refer to the offspring, formed by asexual method of reproduction, a clone?
7. Distinguish between proter and opisthe.
8. How is sexual reproduction advantageous in evolution?
9. Distinguish between lobopodium and filopodium. Give an example to each of them.
10. Define conjugation with reference to ciliates. Give two examples.

SHORT ANSWER TYPE QUESTIONS

1. Name the system that controls the fastest swimming movement of protozoans and write its components.
2. Write the mechanism of bending of flagellum and explain effective and recovery strokes.
3. What are lateral appnedages? Based on their presence and absence, write the various types of flagella giving at least one example for each type.
4. Describe the process of transverse binary fission in *Paramecium*.
5. Describe the process of longitudinal binary fission in *Euglena*.
6. Write a short note on multiple fission.
7. Give an account of pseudopodia.
8. Give an account of the ultra structure of an axoneme.
9. Draw a neat labelled diagram of ***Euglena***.
10. Draw a neat diagram of ***Paramecium*** and label its important structures/ components.



For ignited minds

- NOT FOR EVALUATION

1. If, a row of lateral appendages is added to the flagellum of *Chilomonas*, what type of flagellum does it transform into, hypothetically speaking?
2. What are the structures associated with locomotion that occur in higher animals such as man?
3. What is the primary source of energy for any type of movement, be it in lower organism or higher organism?
4. Why should conjugation occur in ciliates only, why not in other protozoans. **(CLUE)**: In the heteronuclear organisms, the vegetative nucleus divides **amitotically**.
5. If the dynein arms are immobilized, do flagella or cilia or both stop functioning?
6. What are the two structures which fix the doublets of a cilium/flagellum and keep them in position or in other words maintain the integrity of microtubules?
7. There is some indication concerning certain gametes of higher organisms, that at least indicate the origin of higher animals from some simple protists. **Can you explain this concept?**
8. *Euglena* has chromatophores. Yet zoologists consider it an animal. From the few things you have learnt in this lesson about *Euglena*, what features speak in favour of the view of zoologists?
9. A *Paramecium* can produce more daughter paramecia at the end of conjugation. The same *Paramecium* can produce a clone of daughter paramecia through multiplication by binary fission. Is there any difference between the two types of daughter paramecia? Discuss with your teacher and find whether your answer is correct or not?
[CLUE- Production of a clone does not involve pronuclei].
10. All undulipodia (**Hyman's terminology**) do not show undular movement. Do you agree with the statement. If so, how?
11. The total number of microtubules present above the level of pellicle of a flagellum is 20.
12. The total number of microtubules present in the kinetosome is 27 (3×9)
13. The 9 **radial spokes** in an axoneme prevent undue sliding of peripheral doublets.
14. The tubules that connect kinetosome to the plasma membrane and nucleus are called **rootlets**.
15. The flagellate with highest number of flagella is *Trichonympha*.
16. Cilia and flagella arise from kinetosomes, their **progenitors**.
17. Flagella and cilia are considered **homologous** and **analogous organelles**.
18. The dynein arms are the sites of activity of **ATPase**, which releases energy from ATP.
19. **Uniparental** inheritance is involved in asexual reproduction, whereas **biparental** inheritance is involved in sexual reproduction.
20. The first stage in the life of a sexually reproducing organism is a **totipotent** diploid cell, the **zygote**.